MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, THERMOCOUPLE, (FOR MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3, MIL-C-83723 SERIES 3, AND MIL-C-83733 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.

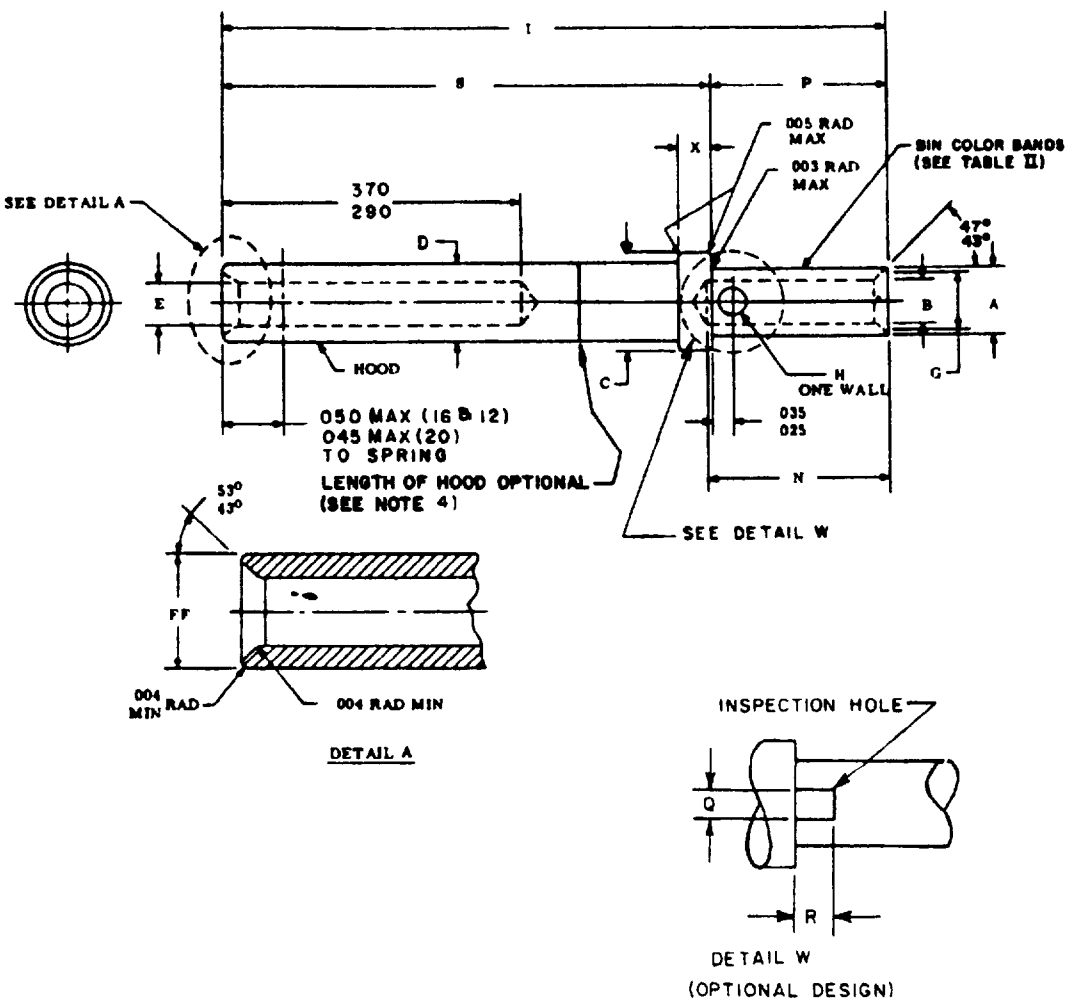


FIGURE 1 Connector contact.

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:30Z
Check the source to verify that this is the current version before use.

INCHES	MM	INCHES	MM
.026	.66	.089	2.26
.029	.69	.100	2.54
.032	.72	.101	2.57
.033	.75	.103	2.62
.036	.79	.110	2.79
.042	.83	.113	2.87
.044	.87	.130	3.30
.048	.92	.133	3.38
.050	.97	.157	3.99
.051	1.02	.160	4.06
.060	1.08	.171	4.34
.062	1.14	.186	4.72
.065	1.21	.246	6.25
.066	1.28	.250	6.35
.068	1.35	.257	6.53
.076	1.43	.284	7.21
.078	1.51	.500	12.70
.083	2.34	.656	16.66
.087	2.43	.759	19.28

TABLE I. Dimensions.

BIN Code	A DIA	B DIA	C DIA	D DIA	E DIA	G DIA	H DIA	L MAX (REF)	N	P	S	X	FF MIN DIA	Q	R
138 through 142	.078	.050	.103	.078	.044	.066	.032	.656	.186	.171	.485	.033	.060	.032	.051
	.076	.048	.100	.076	.042	.062	.026		.157	.160	.479	.029		.026	.026
519 through 523	.103	.068	.133	.113	.068	.089	.042	.759	.284	.257	.500	.048	.087	---	---
	.101	.066	.130	.110	.065	.083	.036		.250	.246	.494	.044			

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Dimensions shown apply after plating.
4. Maximum cap of .010 inch (.25 mm) between hood and body of the contact.

FIGURE 1. Connector contact - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:30Z
Check the source to verify that this is the current version before use.

TABLE II. Design characteristics.

BIN code	Color bands			Mating end size	Wire barrel size	Type	Class
	1st	2nd.	3rd.				
138	Brown	Orange	Gray	20	20	C see table IV	B
139	Brown	Orange	White				
140	Brown	Yellow	Black				
141	Brown	Yellow	Brown				
142	Brown	Yellow	Red				
519	Green	Brown	White	16	16	C see table IV	B
520	Green	Red	Black				
521	Green	Red	Brown				
522	Green	Red	Red				
523	Green	Red	Orange				

TABLE III. Tools.

BIN code	Basic crimping tool	Positioner	Installing tool	Removal tool	
				Wired contact	Unwired contact
138	M22520/2-01 or M22520/1-01	M22520/2-02 Red or M22520/1-02	M81969/14-02 Red	M81969/14-02 White	M81969/30-05 Red
139					
140					
141					
142					
519	M22520/1-01 or M22520/7-01	M22520/1-02 Blue or M22520/7-03	M81969/14-03 Blue	M81969/14-03 White	M81969/30-06 Blue
520					
521					
522					
523					

TABLE IV. Materials, plating, and tensile strength.

BIN code	Materials	Plating	Tensile strength			
			Wire size	Axial load	Wire size	Axial load
138	Copper	Gold Flash	24	6 lbs.	20	12 lbs.
139	Constantan	None				
140	Alumel	None				
141	Chromel	None				
142	Iron	Cadmium Plate				
519	Copper	Gold Flash	20	14 lbs.	16	45 lbs.
520	Constantan	None				
521	Alumel	None				
522	Chromel	None				
523	Iron	Cadmium Plate				

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:30Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions, design characteristics, and configuration See figure 1 and tables I and II.

Tools See table III.

Mating contacts MIL-C-39029/9

Materials and plating See table IV

Vibration Test condition VI, letter J of method 2005 of MIL-STD-1344, except the duration shall be 8 hours in the longitudinal direction and 8 hours in the perpendicular direction (16 hours total).

Shock (specified pulse) Test condition D of method 2004 of MIL-STD-1344, except the pulse shall be an approximate half-sine wave of 3009g ±15 percent magnitude with a duration of 3 milliseconds ±1 One shock shall be exerted in each of three mutually perpendicular directions.

Tensile strength See table IV.

QPL evaluating activity. Naval Avionics Center, Code D/714, 6000 East 21st Street, Indianapolis, Indiana 46218.

International interest. NEPR 57.

Part numbers. See table V.

TABLE V. Part number.

BIN code	Part number	Supersedes	Manufacturer's code
138	M39029/9-138	M39029/10 20 20 C1	N/A
139	M39029/9-139	M39029/10 20 20 C2	
140	M39029/9-140	M39029/10 20 20 C3	
141	M39029/9-141	M39029/10 20 20 C4	
142	M39029/9-142	M39029/10 20 20 C5	
519	M39029/9-519	5000-108-516	14283
520	M39029/9-520	5000-108-416	
521	M39029/9-521	5000-108-116	
522	M39029/9-522	5000-108-216	
523	M39029/9-523	5000-108-316	

Revision letters are not used to denote changes due to the extensiveness of the changes

Custodians
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR, MI
Navy - EC, MC, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Navy - OS

Agent
DLA - ES

Preparing activity
Navy - AS

(Project 5935-3250-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:30Z
Check the source to verify that this is the current version before use.